Chapter Four
The Theory of Production and Cost

4.1 Theory of production in the short run

4.1.1 Definition of production

Transforming raw materials into outputs requires inputs such as **land, labour**, **capital** and **entrepreneurial** ability.

Production is the process of transforming inputs into outputs. It can also be defined as an act of creating value or utility.

4.1.2 Production function

Production function is a technical relationship between inputs and outputs. It shows the maximum output that can be produced with fixed amount of inputs and the existing technology. A production function may take the form of an algebraic equation, table or graph.

A general equation for production function can, for instance, be described as:

$$Q = f(X_1, X_2, X_3, ..., X_n)$$

where, Q is output and X_1, X_2, $X_3, ..., X_n$ are different types of inputs.

Inputs are commonly classified as fixed inputs or variable inputs.

1. Fixed inputs are those inputs whose quantity cannot readily be changed when market conditions indicate that an immediate adjustment in output is required. In fact, no input is ever absolutely fixed but may be fixed during an **immediate requirement**. **For example**, if the demand for Beer rises suddenly in a week, the brewery factories cannot plant additional machinery overnight and respond to the increased demand. Buildings, land and machineries are examples of fixed inputs because their quantity cannot be manipulated easily in a short period of time.

2. **Variable inputs** are those inputs whose quantity can be altered almost instantaneously in response to desired changes in output. That is, their quantities can easily be diminished when the market demand for the product decreases and vice versa. The best example of variable input is unskilled **labour.**

In economics, ***short run*** refers to a period of time in which the quantity of at least one input is fixed.

Consider a firm that uses two inputs: **capital (fixed input**) and **labour** (variable input). Given the assumptions of short run production, the firm can increase output only by increasing the amount of labour it uses. Hence, its production function can be given by:

Q = f (L) where, **Q** is output and L is the quantity of labour.

4.1.3 Total, average, and marginal product

In production, the contribution of a variable input can be described in terms of total, average and marginal product.

Total product (TP): it is the total amount of output that can be produced by efficiently utilizing specific combinations of the variable input and fixed input.

- Increasing the variable input (while some other inputs are fixed) can increase the total product only up to a certain point
- In general, the TP function in the short-run follows a certain trend: **it initially increases at an increasing rate,** then **increases at a decreasing rate**, **reaches a maximum point** and **eventually falls** as the quantity of the variable input rises. This tells us what shape a total product curve assumes.

Marginal Product (MP): it is the change in output attributed to the addition of one unit of the variable input to the production process, other inputs being constant.

For instance, the change in total output resulting from employing additional worker (holding other inputs constant) is the marginal product of labour (*MPL)*.

In other words, *MPL measures the slope of the total product curve at a given point*

$$MP_L = \frac{dTP}{dL} = \frac{\Delta Q}{\Delta L}$$

In the short run, the marginal product of the variable input first increases, reaches its maximum and then decreases to the extent of being negative.

Average Product (AP): Average product of an input is the level of output that each unit of input produces, on the average. It tells us the mean contribution of each variable input to the total product.

Mathematically, it is the ratio of total output to the number of the variable input.

$$AP_L == \frac{TP}{L}$$

Average product of labour first increases, reaches its maximum value and eventually declines.

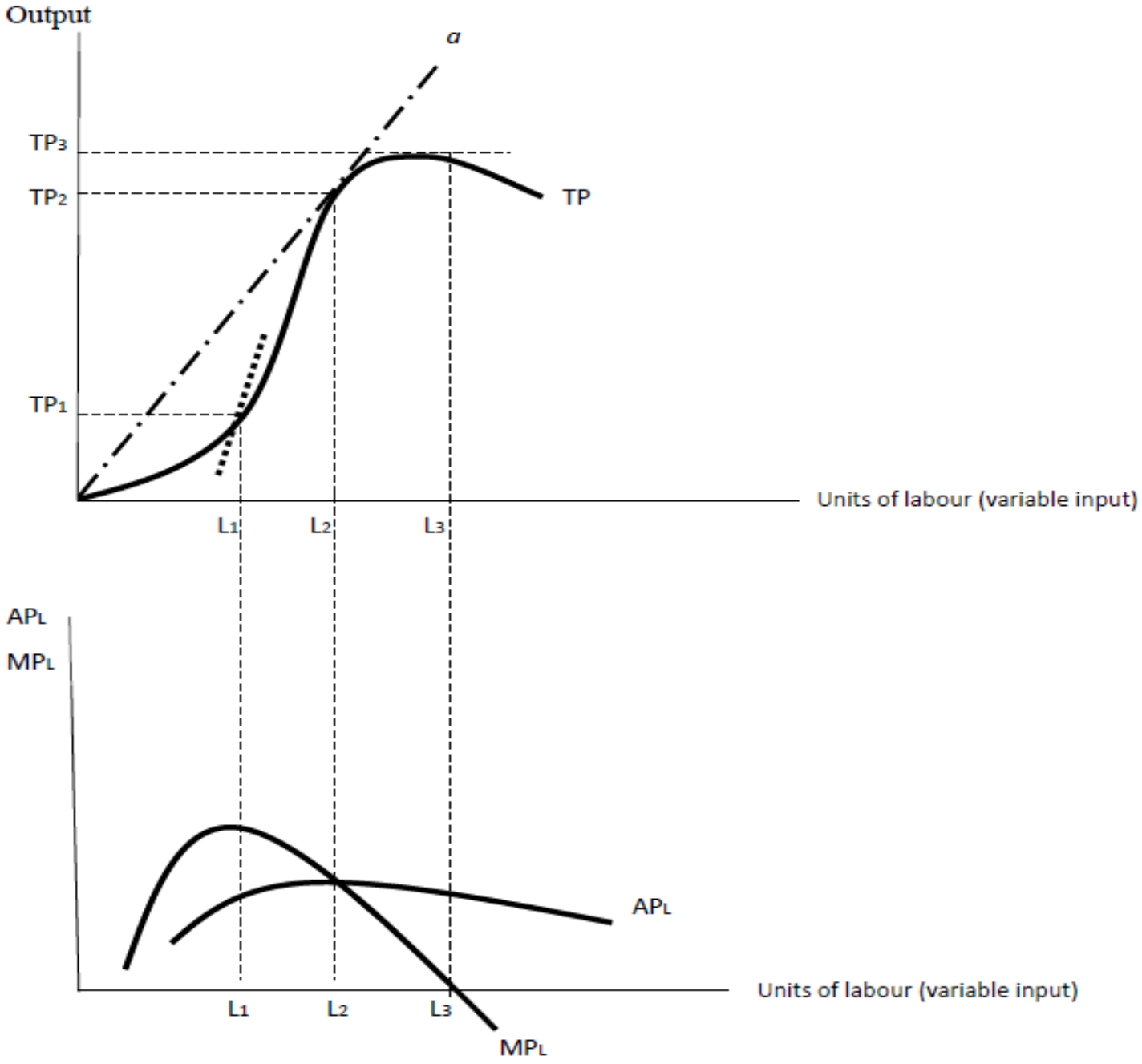


Figure 4.1: Total product, average product and marginal product curves

The relationship between MPL and APL can be stated as follows.

– When APL is increasing, MPL > APL.

– When APL is at its maximum, MPL = APL.

– When APL is decreasing, MPL < APL.

Example: Suppose that the short-run production function of certain cut-flower firm is given by: $Q=4KL-0.6K-0.1L$ where Q is quantity of cut-flower produced, L is labour input and K is fixed capital input (K=5).

a) Determine the average product of labour (APL) function.

b) At what level of labour does the total output of cut-flower reach the maximum?

c) What will be the maximum achievable amount of cut-flower production?

4.1.4 The law of variable proportions

The **law of variable proportions** states that as successive units of a variable input(say, labour) are added to a fixed input (say, capital or land), beyond some point the extra, or marginal, product that can be attributed to each additional unit of the variable resource will decline.

The law starts to operate after the marginal product curve reaches its maximum (this happens when the number of workers exceeds L1 in figure 4.1).

This law is also called the *law of diminishing returns*.

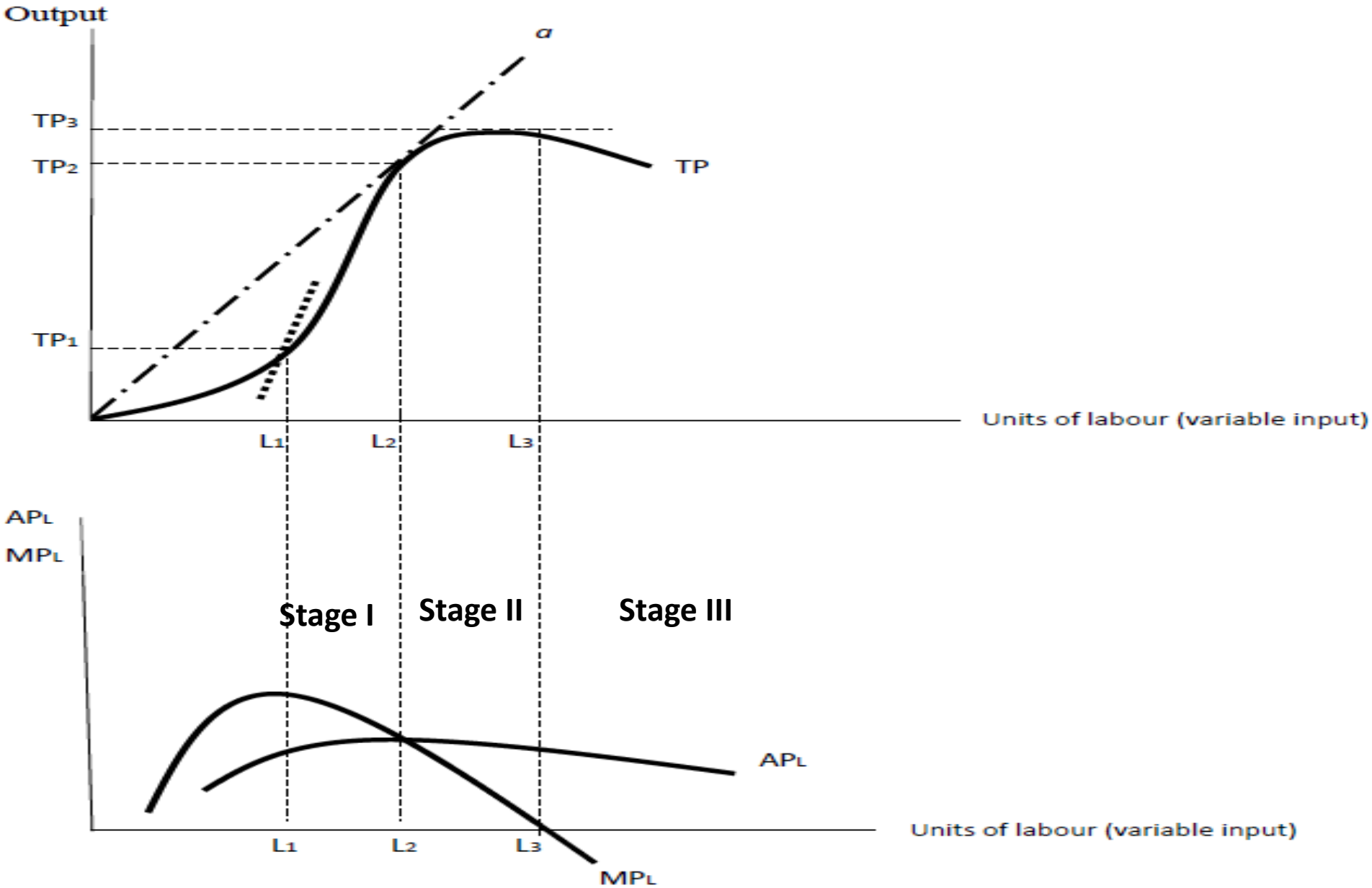


Figure 4.1: Total product, average product and marginal product curves

4.1.5 Stages of production

It is possible to determine the ranges over which the variable input (labour) be employed. To this end, economists have defined three stages of short run production.

Stage I: This stage of production covers the range of variable input levels over which the average product (AP*L*) continues to increase.

It goes from the origin to the point where the APL is maximum, which is the equality of MP*L* and AP*L* (up to L2 level of labour employment in figure 4.1).

This stage is **not an efficient region** of production though the MP of variable input is positive. The reason is that the variable input (the number of workers) is too small to efficiently run the fixed input so that the **fixed input is under-utilized** (not efficiently utilized).

Stage II: It ranges from the point where AP*L* is at its maximum (MP*L*=AP*L*) to the point where MP*L* is zero (from L2 to L3 in figure 4.1).

Here, as the labour input increases by one unit, output still increases but at a decreasing rate.

Due to this, the second stage of production is termed as the stage of ***diminishing marginal returns*****.**

The reason for decreasing average and marginal products is due to the scarcity of the fixed factor.

That is, once the optimum capital-labour combination is achieved, employment of additional unit of the variable input will cause the output to increase at a slower rate. As a result, the marginal product diminishes.

This stage is the efficient region of production. Additional inputs are contributing positively to the total product and MP of successive units of variable input is declining (indicating that the fixed input is being optimally used).

Hence, the efficient region of production is where the **marginal product of the variable input is declining but positive.**

Stage III: In this stage, an increase in the variable input is accompanied by decline in the total product.

Thus, the total product curve slopes downwards, and the marginal product of labour becomes negative.

This stage is also known as the stage of ***negative marginal returns*** to the variable input.

The cause of negative marginal returns is the fact that the volume of the variable inputs is quite excessive relative to the fixed input; the fixed input is **over-utilized**.

Obviously, a rational firm **should not operate in stage III** because additional units of variable input are

contributing negatively to the total product (MP of the variable input is negative). In figure 4.1, this stage is indicated by the employment of labour beyond L3.

4.2 Theory of costs in the short run

4.2.1 Definition and types of costs

Cost is, therefore, the monetary value of inputs used in the production of an item.

Economists use the term —**profit** differently from the way **accountants use** it.

- To the **accountant,** profit is the firm‘s **total revenue** less its **explicit costs** (accounting costs).
- To the **economist**, economic profit is **total revenue** less **economic costs** (explicit and implicit costs).

Accounting cost is the monetary value of all purchased inputs used in production; it ignores the cost of non-purchased (self-owned) inputs. It considers only direct expenses such as wages/salaries, cost of raw materials, depreciation allowances, interest on borrowed funds and utility expenses (electricity, water, telephone, etc.). These costs are said to *be **explicit costs***.

Explicit costs are out of pocket expenses for the purchased inputs. If a producer calculates her

cost by considering only the costs incurred for purchased inputs, then her profit will be an

accounting profit.

Accounting profit = Total revenue – Accounting cost = Total revenue – Explicit cost

In the real world economy, entrepreneurs may use some resources which may not have direct monetary expense since the entrepreneur can own these inputs himself or herself. The estimated monetary cost for non-purchased inputs is known as ***implicit cost*.**

Economic profit =Total revenue – Economic cost (Explicit cost + Implicit cost)

- **Economic profit** will give the real profit of the firm since all costs are taken into account.
- Accounting profit of a firm will be greater than economic profit by the amount of implicit cost.
- If all inputs are purchased from the market, accounting and economic profit will be the same.
- However, if implicit costs exist, then accounting profit will be larger than economic profit.

4.2.2 Total, average and marginal costs in the short run

- A cost function shows the total cost of producing a given level of output. C = f (Q), where C is the total cost of production and Q is the level of output.
- In the **short run**, total cost (TC) can be broken down in to two – total fixed cost (TFC) and total variable cost (TVC). ***TC = TFC + TVC***
- **Total fixed cost (TFC):** Total fixed cost is denoted by a straight line parallel to the output

axis. This is because such costs do not vary with the level of output.

Total variable cost (TVC): cost which vary with level of output. The total variable cost of a firm has an inverse S-shape. The shape indicates the **law of variable proportions in production.**

Total Cost (TC): The total cost curve is obtained by vertically adding TFC and TVC at each level of output. The shape of the TC curve follows the shape of the TVC curve, i.e. the TC has also an inverse S-shape.

It should be noted that when the level of output is zero, TVC is also zero which implies TC = TFC.

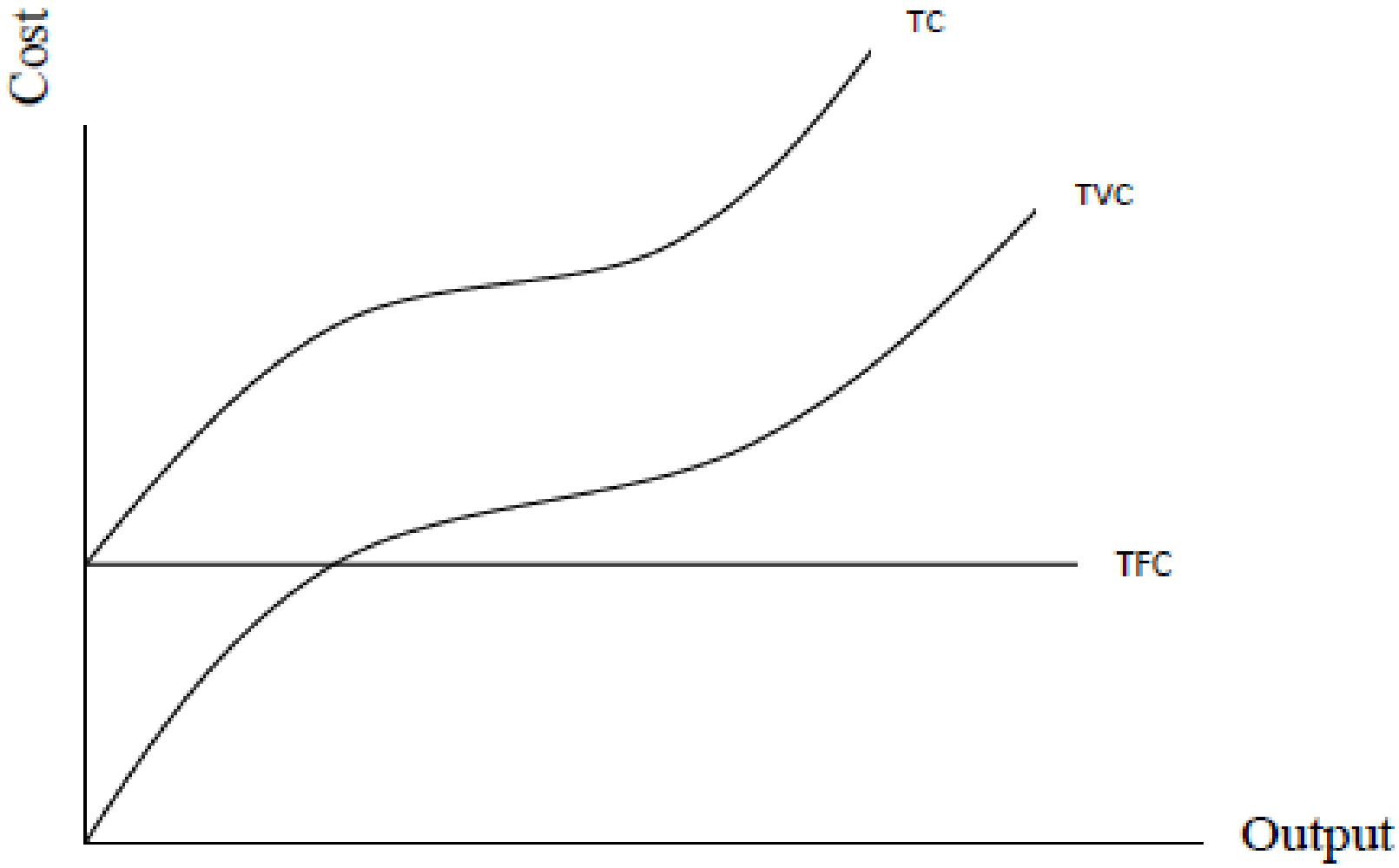


Figure 4.2: ***Short run TC, TFC and TVC curves***

Per unit costs

a) **Average fixed cost (AFC)** - Average fixed cost is total fixed cost per unit of output. $AFC = \frac{TFC}{Q}$

b) Average variable cost (AVC) - Average variable cost is total variable cost per unit of

output. $AVC = \frac{TVC}{Q}$

The short run AVC falls initially, reaches its minimum, and then starts to increase. Hence,

the AVC curve has U-shape and the reason behind is the law of variable proportions.

c) Average total cost (ATC) or simply Average cost (AC) - Average total cost is the total cost per unit of output.

$$AC = \frac{TC}{Q}$$

Equivalently, $AC = \frac{TVC + TFC}{Q} = \frac{TVC}{Q} + \frac{TFC}{Q} = \text{AVC} + \text{AFC}$

Thus, AC can also be given by the vertical sum of AVC and AFC.

Marginal Cost (MC)

Marginal cost is defined as the additional cost that a firm incurs to produce one extra unit of output. In other words, it is the change in total cost which results from a unit change in output. Graphically, MC is the slope of TC function.

$$MC = \frac{dTC}{dQ}$$

In fact, MC is also a change in TVC with respect to a unit change in the level of output.

$$MC = \frac{dTFC + dTVC}{dQ} = \frac{dTVC}{dQ}, \quad \text{since} \quad \frac{dTFC}{dQ} = 0$$

Given inverse S-shaped TC and TVC curves, MC initially decreases, reaches its minimum and then starts to rise. From this, we can infer that the reason for the MC to exhibit U shape is also the law of variable proportions. In summary, AVC, AC and MC curves are all U-shaped due to the law of variable proportions.

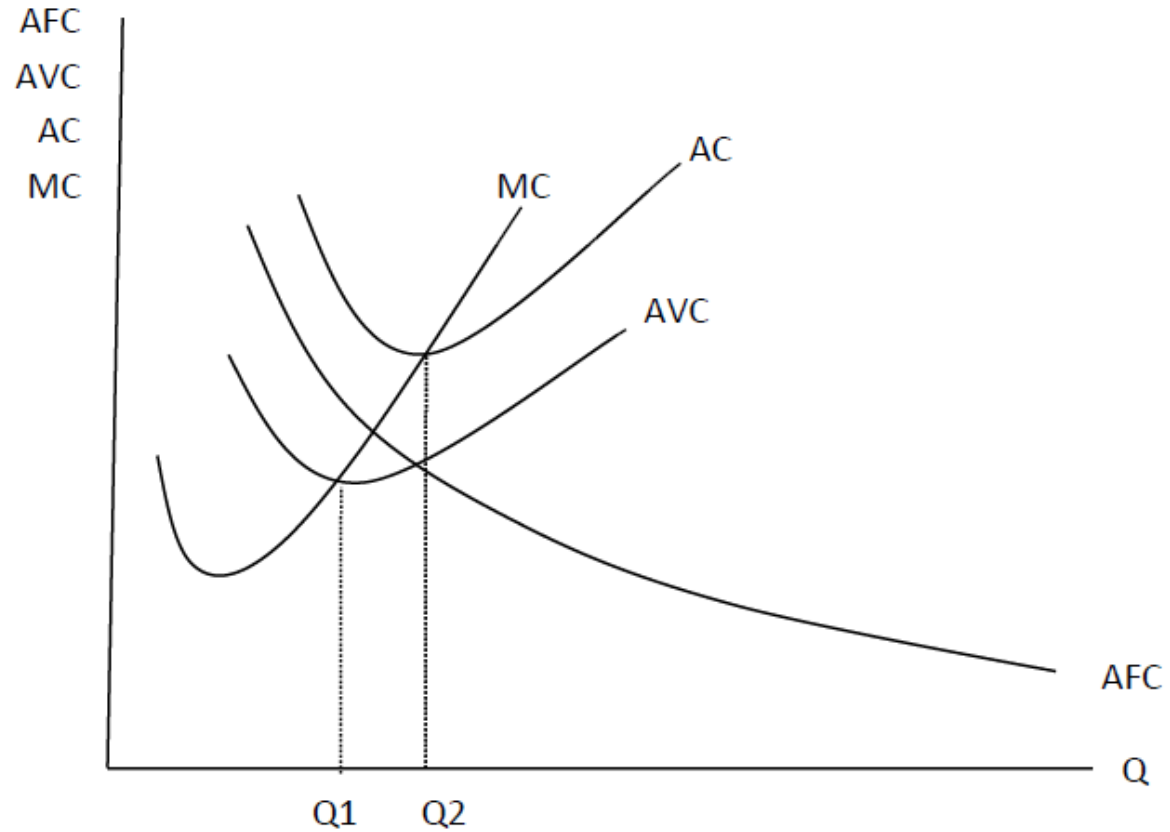


Figure 4.3*: Short run AFC, AVC, AC and MC Curves*

In the above figure, the AVC curve reaches its minimum point at Q_1 level of output and AC reaches its minimum point at Q_2 level of output. The vertical distance between AC and AVC, that is, AFC decreases continuously as output increases. It can also be noted that the MC curve passes through the minimum points of both AVC and AC curves.

Example: Suppose the short run cost function of a firm is given by: $TC=2Q^3-2Q^2+Q+10$.

a) Find the expression of TFC & TVC

b) Derive the expressions of AFC, AVC, AC and MC

c) Find the levels of output that minimize MC and AVC and then find the minimum values of MC and AVC

Solution:

Given $TC=2Q^3-2Q^2+Q+10$

a) $TFC = 10$, $TVC = 2Q^3-2Q^2+Q$

b) $AFC = TFC/Q = 10/Q$

$AVC = TVC/Q = (2Q^3-2Q^2+Q)/Q = 2Q^2-2Q+1$

$AC = TC/Q = (2Q^3-2Q^2+Q+10)/Q = 2Q^2-2Q+1+10/Q$

$MC = dC/dQ = 6Q^2-4Q+1$

c) To find the minimum value of MC,

$$dMC/dQ = 12Q - 4 = 0$$

$$Q = 1/3$$

MC is minimized when $Q = 0.33$

The minimum value of MC will be:

$$MC = 6Q^2 - 4Q + 1$$

$$= 6(1/3)^2 - 4(1/3) + 1 = 0.33$$

To find the minimum value of AVC

$$dAVC/dQ = 4Q - 2 = 0$$

$$Q = 0.5$$

AVC is minimized at Q = 0.5

The minimum value of AVC will be:

$$AVC = 2Q^2 - 2Q + 1$$

$$AVC = 2\ (0.5)^2 - 2(0.5) + 1$$

$$= 0.5 - 1 + 1$$

$$= 0.5$$

4.2.3 The relationship between short run production and cost curves

Suppose a firm in the short run uses labour as a variable input and capital as a fixed input. Let the price of labour be given by w, which is constant. Given these conditions, we can derive the relation between MC and MP_L as well as the relation between AVC and AP_L.

i) Marginal Cost and Marginal Product of Labour

$$MC = \frac{\Delta TVC}{\Delta Q}, \text{where } TVC = w.L$$

$$MC = \frac{\Delta(w.L)}{\Delta Q} = w.\frac{\Delta L}{\Delta Q}, \text{ but } \frac{\Delta L}{\Delta Q} = \frac{1}{MP_L}$$

Therefore, $MC = \frac{w}{MP_L}$

The above expression shows that MC and MP_L are inversely related. When initially MP_L increases, MC decreases; when MP_L is at its maximum, MC must be at a minimum and when finally MP_L declines, MC increases.

ii) Average Variable Cost and Average Product of Labour

$$AVC = \frac{TVC}{Q}, \text{ where, } TVC = w.L$$

$$AVC = \frac{w.L}{Q} = w.\frac{L}{Q}, \text{ but } \frac{L}{Q} = \frac{1}{AP_L}$$

Therefore, $AVC = \frac{w}{AP_L}$

This expression also shows inverse relation between AVC and AP_L. When AP_L increases, AVC decreases; when AP_L is at a maximum, AVC is at a minimum and when finally AP_L declines, AVC increases.

We can also sketch the relationship between these production and cost curves using graphs.

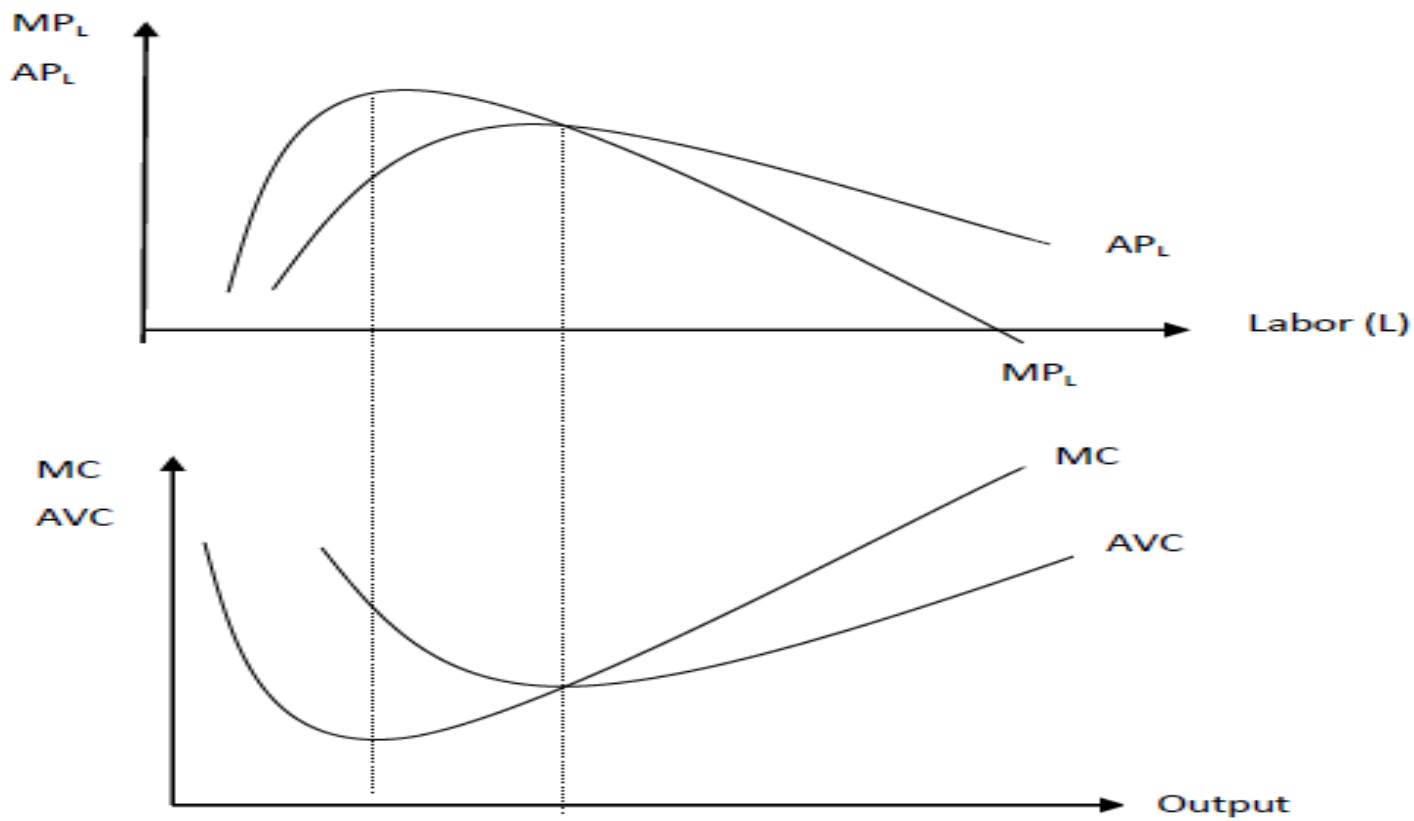


Figure 4.4: relationship between short run production and cost curves

From the above figure, we can conclude that the MC curve is the mirror image of MP_L curve and AVC curve is the mirror image of AP_L curve.